Practice workbook · Explain red-cell adaptations and the effects of more or less oxygen carriage

How red-cell structure suits oxygen transport, and what follows when haemoglobin, red-cell number or oxygenation changes.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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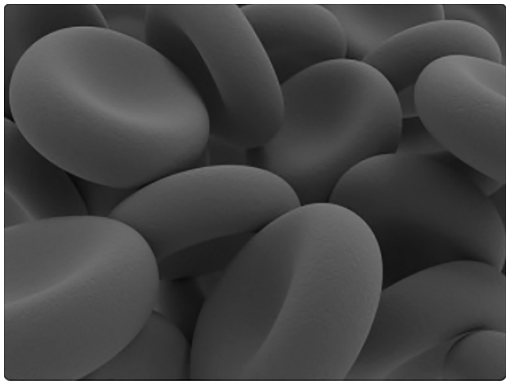
Type ID: HBI-EXPLAIN-OXYGEN-CARRIAGE-AND-ITS-CHANGES

# 1. Question 1 · part a.ii · 4 marks

November 2021 Paper 2

Worked-example retry · Attempt part a.ii only.

**1** (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

**(3)**

(ii) Explain how the structure of the red blood cell is adapted to its function.

**(4)**

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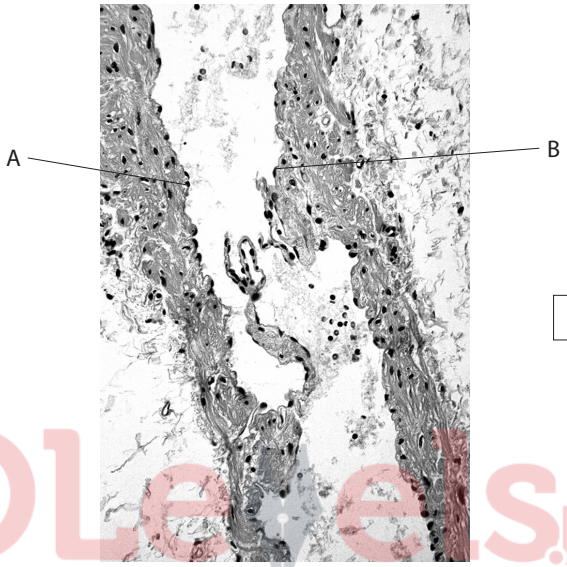
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# 1. Question 1 · part a.ii · 4 marks

November 2021 Paper 2

Worked-example retry · Attempt part a.ii only.

(b) The photograph shows a section through a blood vessel in a leg.




magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

**(1)**

(ii) Explain your choice of blood vessel.

**(2)**

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## 1. Question 1 · part a.ii · 4 marks · Working and answer

Working and answer

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## 2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

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**5** Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

**(2)**

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

**(2)**

.....

.....

.....

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## 2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

| Period of time in hours | Percentage change in mass of haemoglobin |
|---|---|
| 0 | 0 |
| 100 | 0.7 |
| 200 | 2.4 |
| 300 | 4.1 |
| 400 | 5.9 |
| 500 | 6.5 |
| 600 | 7.2 |

(i) Plot the results on the grid.

**(4)**

(ii) Draw the curve of best fit.

**(1)**

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## 2. Question 5 · part b.iii · 5 marks

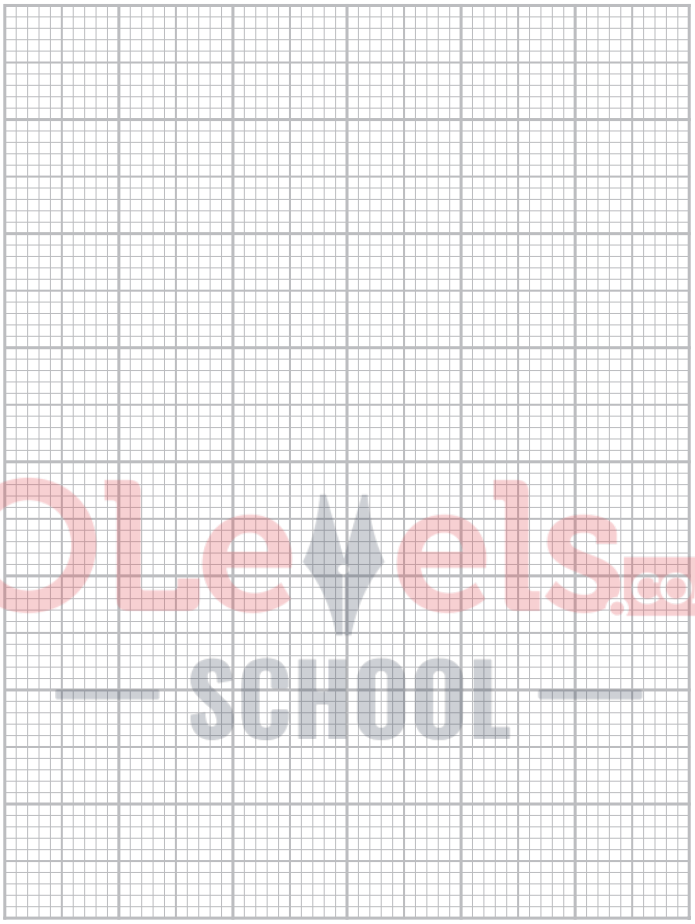
January 2023 Paper 2

Further practice · Attempt part b.iii only.

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## 2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

(iii) Discuss how the percentage change in mass of haemoglobin, after training at altitude, could result in an increase in performance during exercise.

**(5)**

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**(Total for Question 5 = 14 marks)**

## 2. Question 5 · part b.iii · 5 marks · Working and answer

Working and answer

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## 3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

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**6** Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

**(4)**

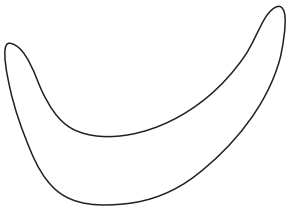
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## 3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

**(5)**

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## 3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

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(b) A woman, heterozygous for sickle cell anaemia, and a man of unknown genotype for sickle cell anaemia have a child. The child is born with sickle cell anaemia.

Explain the possible genotypes of the man.

Use H to represent the allele for unaffected haemoglobin and h to represent the allele for sickle haemoglobin.

**(4)**

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**(Total for Question 6 = 13 marks)**

## 3. Question 6 · part a.ii · 5 marks · Working and answer

Working and answer

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# 4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

**5** (a) The table gives information about some blood components in three people.

| Blood component | Healthy person number per $cm^3$ | Person A number per $cm^3$ | Person B number per $cm^3$ |
|---|---|---|---|
| white blood cells | 7500 | 13 750 | 7250 |
| red blood cells | $5.0 \times 10^6$ | $5.0 \times 10^6$ | $5.0 \times 10^6$ |
| platelets | $3.0 \times 10^5$ | $3.0 \times 10^5$ | $0.8 \times 10^5$ |

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

**(2)**

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

**(2)**

.....

.....

.....

.....

(iii) Describe the effect that the lower number of platelets per $cm^3$ would have on the health of person B.

**(2)**

.....

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P 7 5 9 5 8 A 0 1 2 2 4

# 4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

**(4)**

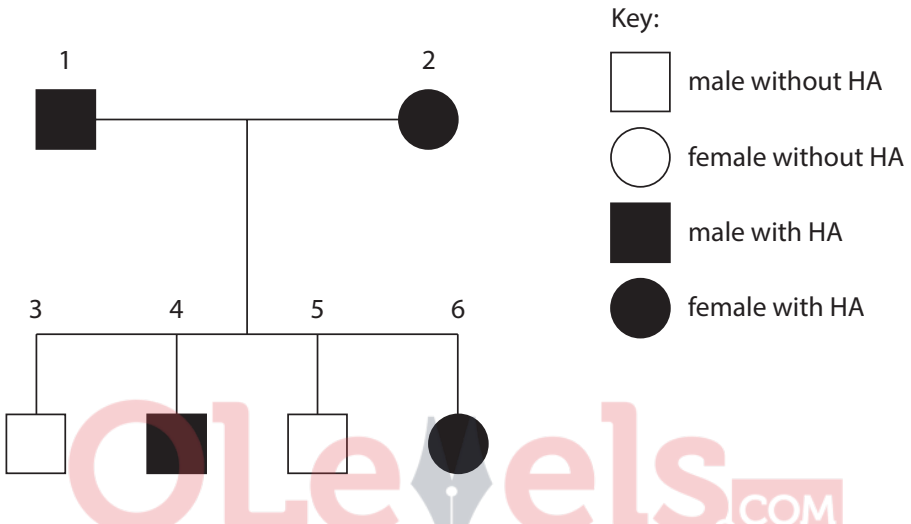
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## 4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

(ii) The diagram shows the inheritance of HA in one family.




Explain how HA is inherited.

**(4)**

**(Total for Question 5 = 14 marks)**

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## 4. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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## 5. Question 4 · part b · 4 marks

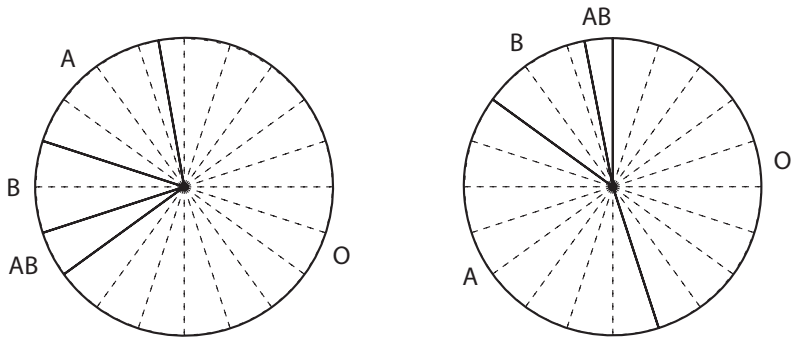
June 2022 Paper 1

Further practice · Attempt part b only.

**4** (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.




(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

**(2)**

**Africa**

| Blood group | Percentage of blood group |
|---|---|
| O | 68 |
| A | |
| B | |
| AB | 5 |

**Europe**

| Blood group | Percentage of blood group |
|---|---|
| O | 45 |
| A | |
| B | 12 |
| AB | |

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# 5. Question 4 · part b · 4 marks

June 2022 Paper 1

Further practice · Attempt part b only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

**(3)**

(iii) A male with the genotype $I^AI^O$ and a female with the genotype $I^BI^O$ have a child with the genotype $I^OI^O$.

Draw a genetic diagram to show how the child inherits the genotype $I^OI^O$.

**(3)**

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

**(1)**

probability = .....

## 5. Question 4 · part b · 4 marks

June 2022 Paper 1

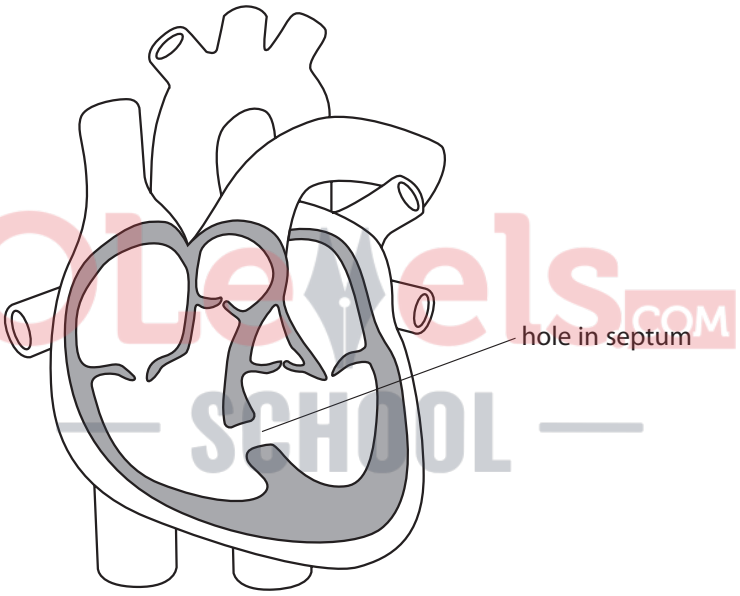
Further practice · Attempt part b only.

(b) The heart pumps the blood around the body.

The diagram shows a human heart.

This heart has a hole in the septum.

This means that blood from the left side of the heart mixes with blood from the right side.




Explain why a person with a hole in their heart breathes at a greater rate than a person with a healthy heart.

**(4)**

**(Total for Question 4 = 13 marks)**

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## 5. Question 4 · part b · 4 marks · Working and answer

Working and answer

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